Using molecular markers to determine the genetic relationships of medicinal plants

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Medicinal plants, primers, PCR amplification, DNA sequences, genetic analysis

ABSTRACT

In developed countries, the use of medicinal plants is currently drawing much attention, as essential constituents of complementary and alternative medications. In order to remove uncertainties regarding their safety and effectiveness, these plants need to be accurately identified and validated. In this regard, genetics has come to the fore as a dependable and quick method for the discerning of medicinal plant sources. This work was a DNA barcoding study, involving genetic and sequence investigates, in addition to a pair of ITS markers, conducted from five families for a total of nine plant varieties. The aim was to reveal the way in which, the four identified DNA barcoding areas of markers (ITS), facilitated the identification and conservation process, and to scrutinise their efficiency in DNA barcoding. The results showed that four sequences of DNA, for the plants under study, as *Carissa macrocarpa*, *Alcea rosea, Delonix regia*, and *Roystonea regia*. Therefore, it can be assumed that using DNA molecular is definitive, when it comes to discerning the genetic links in medicinal plants. From assessing population configurations and genetic diversity to phylogenetic research, facilitating reproduction programmes, and quality control activities, the application of genetic analysis is both wide-ranging and effective. The assessment of DNA markers, a protocol with conventional expertise, will serve to improve our information and knowledge of medicinal plants, while promoting their sustainable usage.

Introduction

Medicinal plants, which are often classified as the basis of modern medicine, have been used since ancient times. Salmerón-Manzano et al. (2020) reported that for many present-day drugs, compounds originating from plants represent important and indispensable components. Alhasnawi (2023) found that the conservation of medicinal plant sources is important, as they are frequently identified as the source for the development of new drugs.

Medicinal plants are used in ethnomedicine and traditional medicine, as well as the treatment for different ailments (Sheidai et al. 2019), more than 1300 medicinal plants are used in European countries. The World Wildlife Fund (WWF), and International Union for Conservation of Nature (IUCN) observed that between 50,000 – 80,000 flowering plants are taken advantage of for their therapeutic properties (Hao 2019).

Recent progress in DNA-based markers has changed our understading of the genetic basis of life, encouraging researchers to strive for a more comprehensive knowledge of the complex genetic analysis contributing towards organism diversity (Durmaz et al. 2015). These advancements have facilitated a more comprehensive and detailed probe into their genetic association with various plants (Beckmann and Lew 2016). Molecular biology is precise DNA sequences, which can be inspected and perceived, to identify the intricate genetic links between the different varieties of medicinal plants. Molecular markers can be used to identify genetic diversity in plants (Alhasnawi 2023). The increasing confidence in natural medications can be attributed to the fewer side effects associated with their usage (Kumar et al. 2017). One of the main reasons hindering the

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promotion of medicinal plant sources usage include the difficulty in authenticating small sample quantities. However, it is essential that an effective method, be used, to accurately determine the molecular markers of medicinal plants. DNA-based markers, an effective means for the determination of genetic diversity assessment in plants, are also employed for the corroboration of herbal remedies, resolving forensic science issues, and food items, and for other uses (Sahare and Srinivasu 2012). This paper explores the relationships and identifies similarities between nine medicinal plants: *Carissa macrocarpa, Alcea rosea, Cichorium intybbus, Silybum marianum, Seaforthia elegans, Delonix regia, Brahea armata, Roystonea regia,* and *Portulacaria afra,* using molecular markers.

Materials and methods

Sample collection and preparation

Table 1 shows the nine plant varieties that were studied. The medicinal plant accumulation process was based on tissue type and plant type. The seeds of *Portulacaria afra, Roystonea regia, Brahea armata, Delonix regia*, and *Seaforthia elegans*, which are plants in the shrub and tree categories, were commercially acquired. *Silybum marianum* and *Cichorium intybbus* leaves were acquired from Egypt seeds, while *Alcea rosea* and *Carissa macrocarpa* leaves were acquired from local plant sale outlets. The accumulation process for *Silybum marianum, Cichorium intybbus*, and *Alcea rosea,* which are categorised as herbs, entailed the collection of their leaves. As for *Carissa macrocarpa,* a shrub from the Apocynaceae family, the accumulation of its leaves is in accordance with its tissue type.

Table 1: The plants used in the study with their family, tissue type and plant type

Plant name	Family	Tissue type used	Plant type
Portulacaria afra	Didiereaceae	Seeds	Shrub
Roystonea regia	Arecaceae	Seeds	Tree
Brahea armata	Arecaceae	Seeds	Tree
Delonix regia	Fabaceae	Seeds	Tree
Seaforthia elegans	Arecaceae	Seeds	Tree
Silybum marianum	Asteraceae	Leaves	Herb
Cichorium intybbus	Asteraceae	Leaves	Herb
Alcea rosea	Malvaceae	Leaves	Herb
Carissa macrocarpa	Apocynaceae	Leaves	Shrub

DNA extraction

The Easy Pure® Plant Genomic DNA Kit (TransGen Biotech, Beijing, China) with specific alterations with regards to the protocols set by the manufacturer, was applied for the DNA extraction process. Pre-heating of the re-suspension buffer 1 (RB1) in a 65 ^{o}C water bath, was followed by the addition of powdered tissue and 130 mg of fresh leaf. Additionally, the tissue lysis incubation time was increased from 15 minutes to 20 minutes, with interval vortexing carried out thrice during the incubation period determined according to protocols set by the manufacturer with some modifications. As depicted in Figure 1, in the final phase, the elution buffer was decreased from 200 to 70 µl. Gel electrophoresis for PCR product, which was visualized under UV light after staining with ethidium bromide lane nine medicinal plants 1; *Portulacaria afra,* 2; *Roystonea regia,* 3; *Brahea armata,* 4; *Delonix regia,* 5; *Seaforthia elegans,* 6; *Silybum marianum,* 7; *Cichorium intybbus,* 8; *Alcea rosea,* 9; *Carissa macrocarpa*.

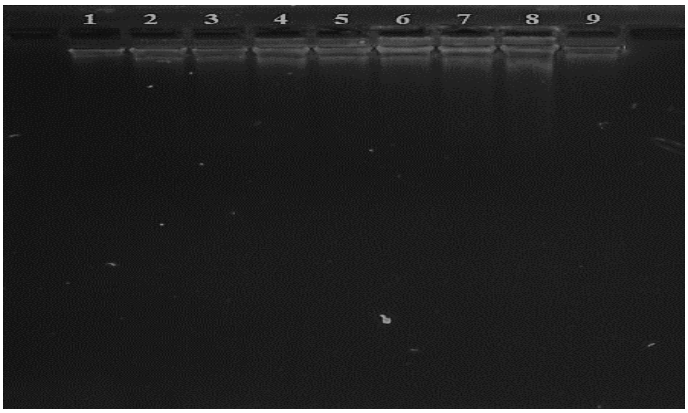


Figure 1: Agarose gel electrophoresis stained with ethidium bromide dye showed a clear band representing DNA molecule that extracted from plant source

PCR amplification

Preparing the primer for the polymerase chain reaction (PCR)

The polymerase chain reaction (PCR) was performed in a 50 µl reaction solution comprising 50 ng of a DNA template, 5 µl of a 10X buffer, 3 µl 25 mM of magnesium chloride ($MgCl_2$), 4 µl 10 mM of deoxynucleotide triphosphates (dNTP), 10 pmol of primers, and 3 units/µl of Eppendorff® Taq polymerase thermostable DNA polymerase I. The use of Sigma Aldrich®'s universal internal transcribed spacer (ITS) primers: Reverse (5'TCCTCCG CTTATTGATATGC-3') and Forward (5'-GGAAGGAGAAGTCGTAACAAGG-3'). 700 bp (Sahare and Srinivasu 2012), was recorded as the PCR product. The solution was put through a session of rapid rotation, and subsequently left to cool overnight at 4°C. The preparation of the working mix involved the dilution of the stock by way of a Tris-EDTA (TE) buffer, to acquire the active mixture of each primer (10 pmol/ml).

Synthesising the master mix for the polymerase chain reaction

Adhering to the manufacturer's instructions, the OneTaq® Quick-Load® 2X Master Mix with Standard Buffer (M0486) was employed to synthesise the PCR master mix reaction. This was followed by the placing of the PCR master mix components into regular PCR tubes holding lyophilised multiplex PCR and other materials essential for the analysis, specifically, 6 mM of $MgCl_2$, taq polymerase thermostable DNA polymerase I, pH 8.7, and dNTP. The tubes were subjected to 3 minutes of rapid rotation, following which, a Labnet MultiGene™ OptiMax thermal cycler (NJ, USA) was employed to process the mixture.

Amplifying the polymerase chain reaction and assessing the DNA

The standard PCR thermocycler technique was applied for every gene, in compliance with the terms specified (Sahare and Srinivasu 2012). The research goal and primer choice are itemised in Table 2.

Table 2: BLAST results found in UniProtKB

Entry	Entry name	Protein names	Gene names	Organism	Length	BLAST
A0A6P5WYZ9	A0A6P5WYZ9_DURZI	Uncharacterised protein LOC111278348	LOC111278348	*Durio zibethinus* (Durian)	269 AA	BLAST 250 results found in UniProtKB
A0A2G2UY24	A0A2G2UY24_CAPBA	Uncharacterised protein	CQW23_34715	*Capsicum baccatum* (Peruvian pepper)	182 AA	BLAST 240 results found in UniProtKB
A0A2G2UZ48	A0A2G2UZ48_CAPBA	Uncharacterised protein	CQW23_34472	*Capsicum baccatum* (Peruvian pepper)	172 AA	BLAST 250 results found in UniProtKB
A0A0J7YM11	A0A0J7YM11_BETVV	Uncharacterised protein	BVRB_017900	*Beta vulgaris subsp. vulgaris* (Beet)	141 AA	BLAST 250 results found in UniProtKB

Separating the DNA fragments using gel electrophoresis

Preparing the agarose gel (1%)

A solution comprising a mixture 1 g agarose and 100 ml 1X TBE buffer was prepared in a flask. The solution was placed in a microwave oven for heating until boiling occurred. The flask was then taken out of the microwave oven, the solution subjected to stirring, and subsequently re-introduced into the microwave oven, until boiling was again evident. This sequence was replicated until all flicks were eradicated. Following the cooling of the agarose at a temperature between 50 - 65°C the resulting agarose gel was merged with ethidium bromide (5 µl solution consisting of 10 mg/ml stock for every 100 ml agarose gel).

Conducting the gel electrophoresis test

The electrophoresis procedure, using agarose gel in five steps, was employed for an assessment of the multi-gene PCR substances. A 1X TBE buffer was subjected to mixing in a water bath for a period of 15 minutes, at a temperature of 100°C and then cooled down to a temperature of 50°C, to synthesise the 1% agarose gel sample. This was followed by the merging of the solution with 2 µL ethidium bromide ($C_{21}H_{20}BrN_3$), and the setting of the gel solution in a tray with the comb affixed in the mandatory position. Cooling of the solution at room temperature, for a period of 15 minutes, led to its solidification. The comb was then gradually removed, and 10 µl of PCR product, together with 5 µl of 1500 bp DNA Ladder, was introduced into single wells. A tray was installed in the electrophoresis structure, and 1X TBE buffer was added to the solution. The processing of the solution involved the use of 100 volts for a period of 30 minutes, and 50 volts for a period of 45 minutes. Ultimately, a graphic representation of the PCR samples was achieved, through the employment of an ultraviolet trans-illumination instrument.

Scoring the data by analysing the banding

The scoring for the ITS primer was based on the presence (1) or absence (0) of the band on the gel. The distance tree of results was computed for all accessions, in compliance with the National Centre for Biotechnology Information (NCBI), and the Basic Local Alignment Search Tool (BLAST) parameters for protein sequence from

sequence similarity, in compliance with the UniProt Knowledge base (UniProtKB).

Results

Figure 2 shows the genomic DNA of the nine plants: *Portulacaria afra, Roystonea regia, Brahea armata, Delonix regia, Seaforthia elegans, Silybum marianum, Cichorium intybbus, Alcea rosea,* and *Carissa macrocarpa.* The findings identify *Roystonea regia, Delonix regia, Alcea rosea,* and *Carissa macrocarpa* as medicinal plants, and *Portulacaria afra, Brahea armata, Seaforthia elegans, Silybum marianum* and *Cichorium intybbus* as non-medicinal plants.

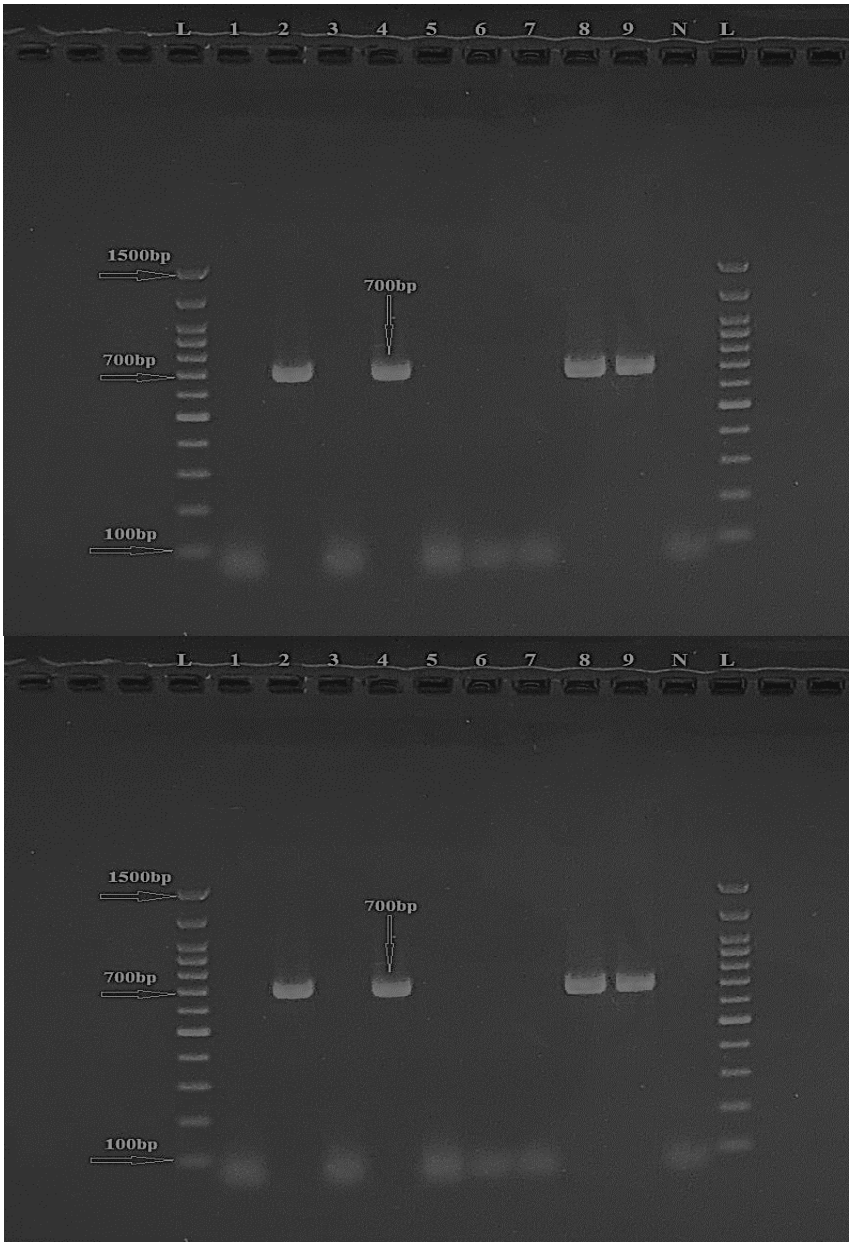
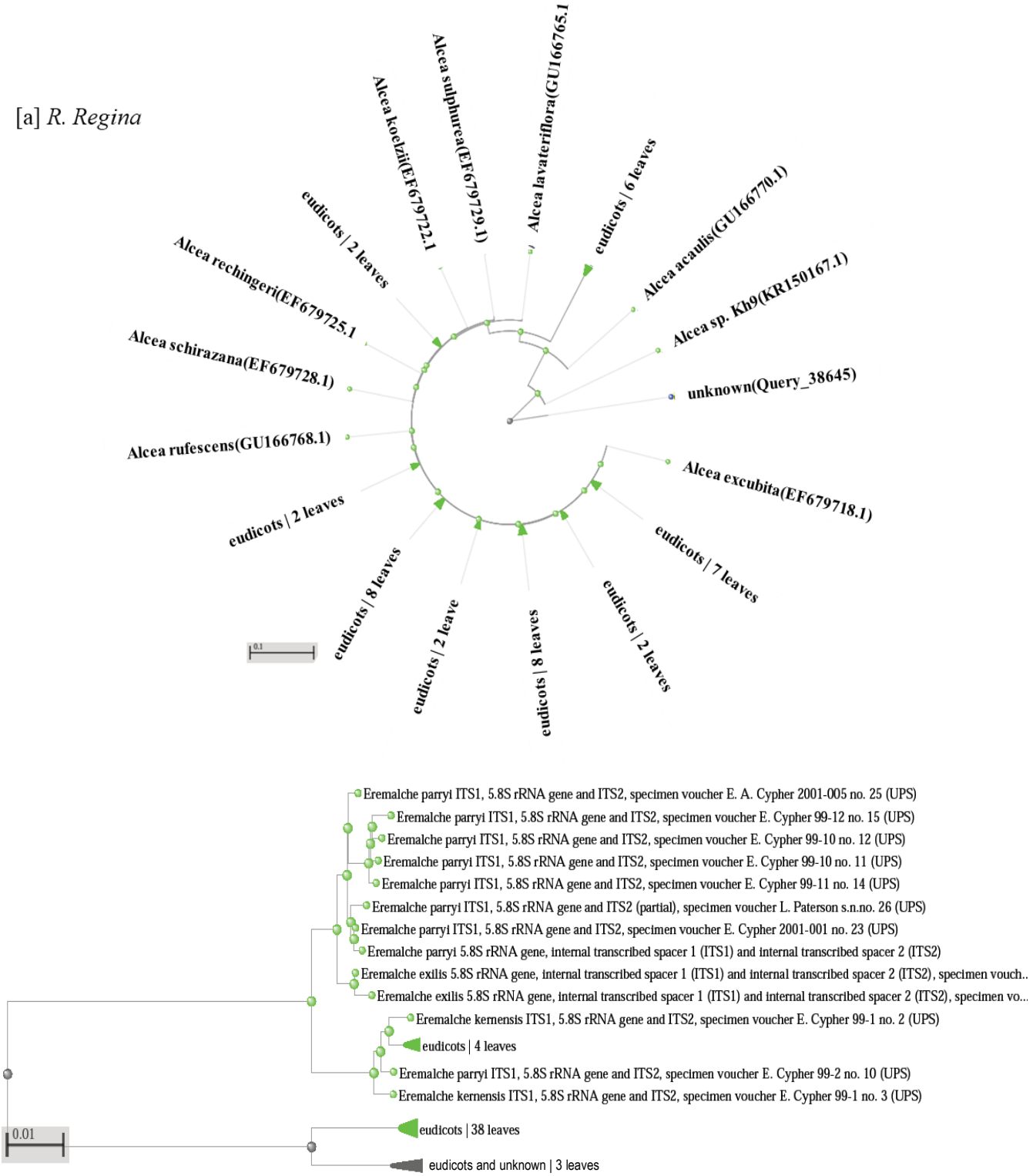
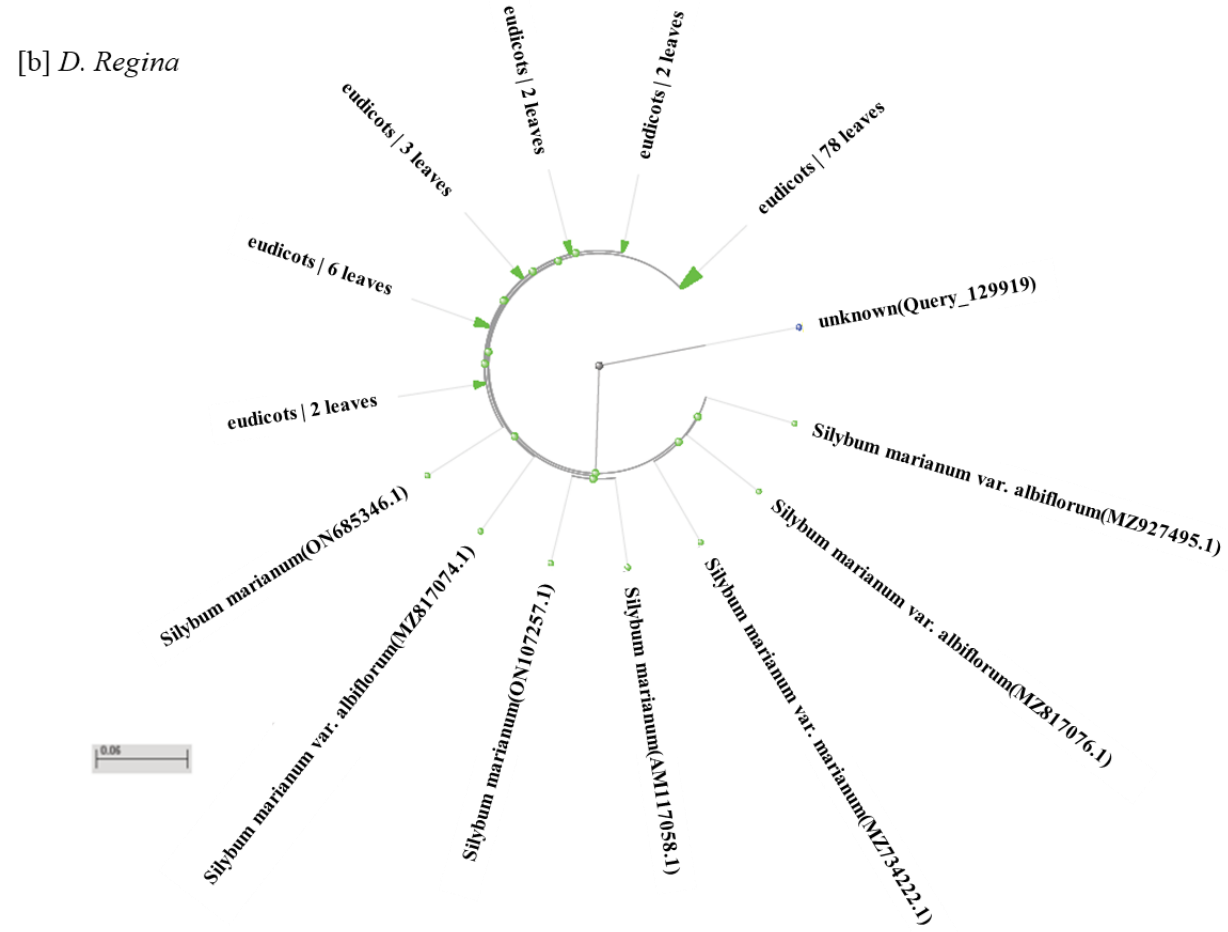


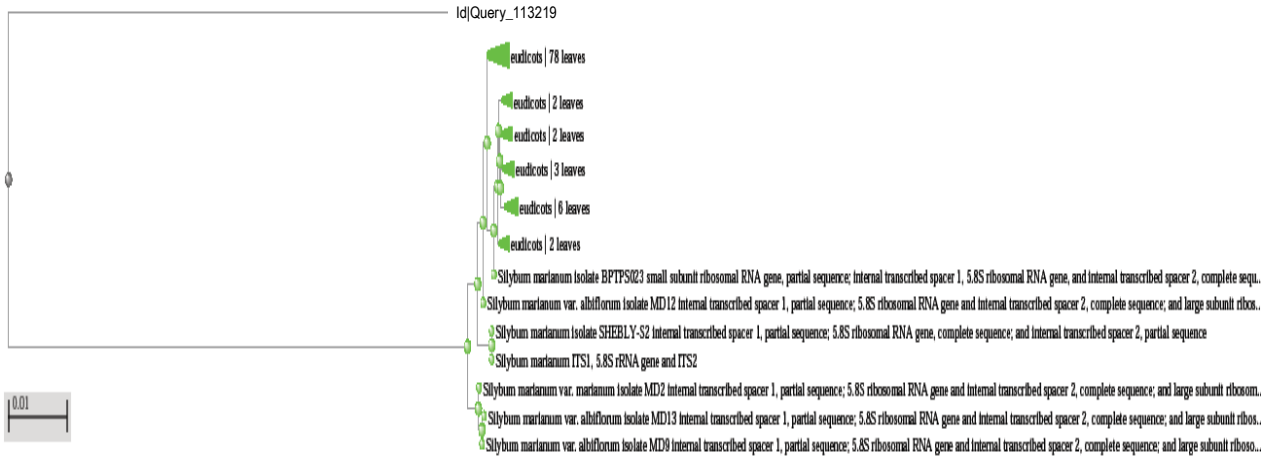
Figure 2: Gel electrophoresis for PCR product of (its universal primer) which show approximately 700 bp at 57 c, (agarose 1.5 %, 10 min at 100 volts and then lowered to 75 volts for 60 min) visualised under UV light after staining with ethidium bromide lane I: DNA ladder (1500-100) bp. Lanes 2: *Roystonea regia*; 4: *Delonix regia*; 8: *Alcea rosea*; 9: *Carissa macrocarpa* represent positive results. Lanes 1: *Portulacaria afra*; 3: *Brahea armata*; 5: *Seaforthia elegans*; 6: *Silybum marianum*; 7: *Cichorium intybbus* represent negative results. Lane N represents negative control.

The distance trees of DNA ITS sequences of *Roystonea regia*, *Delonix regia*, *Alcea rosea*, and *Carissa macrocarpa* display notable alignments using Molecular Evolutionary Genetics Analysis (MEGA) software, Version 11.0.13. (Figure 3)

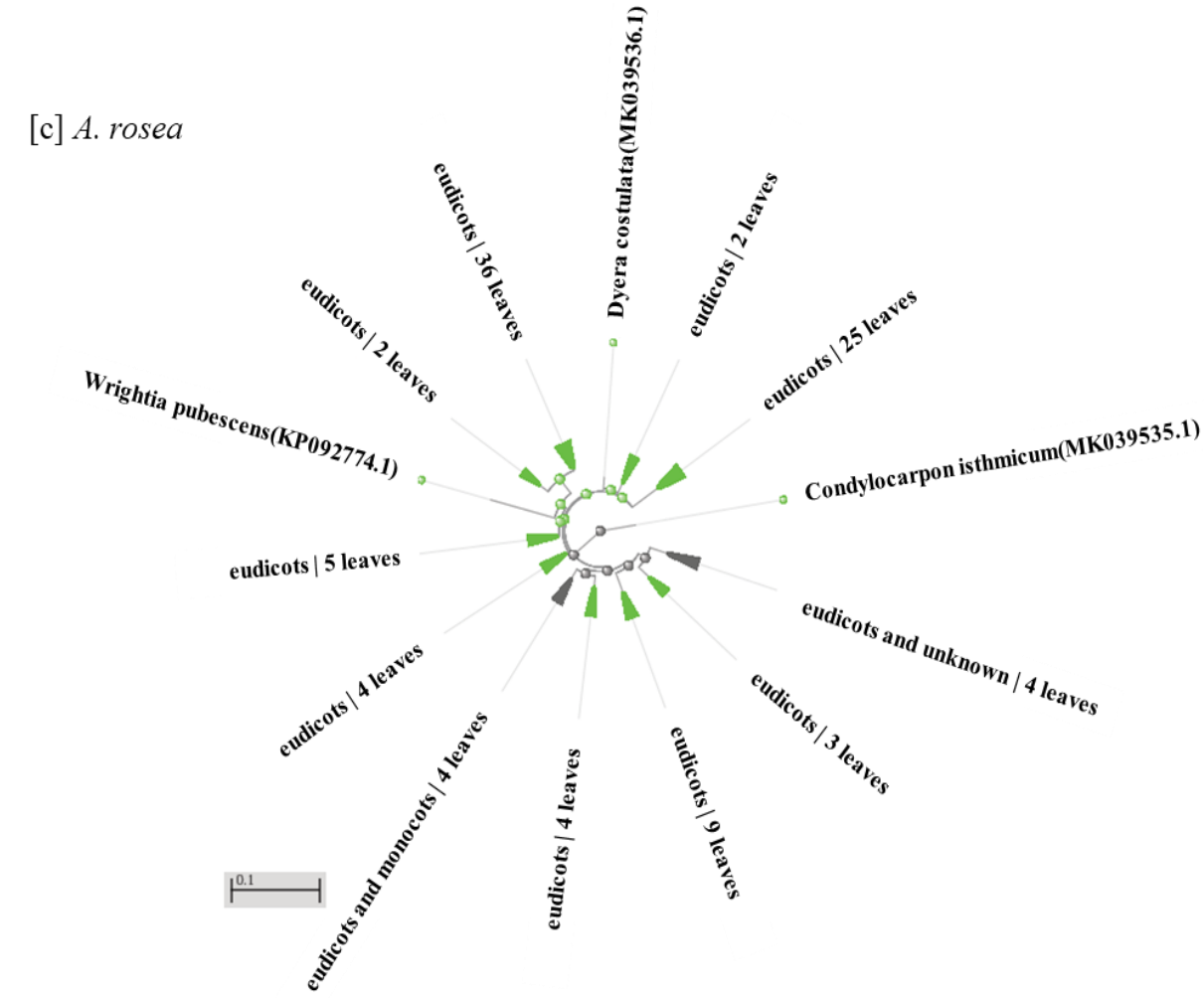


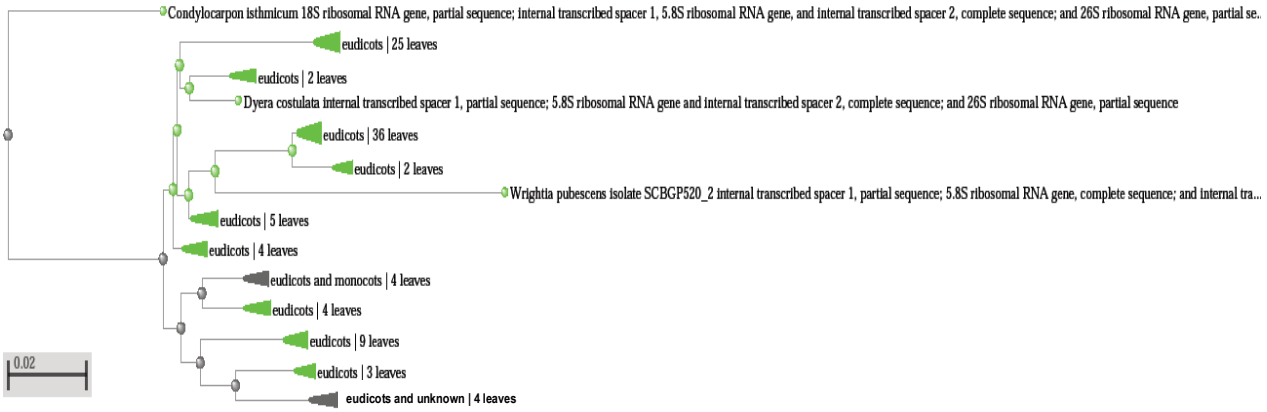
[b] *D. Regina*


eudicots | 2 leaves
eudicots | 2 leaves
eudicots | 3 leaves
eudicots | 78 leaves
eudicots | 6 leaves
unknown(Query_129919)
eudicots | 2 leaves
Silybum marianum var. albiflorum(MZ927495.1)
Silybum marianum(ON685346.1)
Silybum marianum var. albiflorum(MZ817076.1)
Silybum marianum var. albiflorum(MZ817074.1)
Silybum marianum(ON107257.1)
Silybum marianum(AM117058.1)
Silybum marianum var. marianum(MZ734222.1)
0.06


Id|Query_113219
eudicots | 78 leaves
eudicots | 2 leaves
eudicots | 2 leaves
eudicots | 3 leaves
eudicots | 6 leaves
eudicots | 2 leaves
Silybum marianum isolate BPTPS023 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequ..
Silybum marianum var. albiflorum isolate MD12 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribos..
Silybum marianum isolate SHEBLY-S2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence
Silybum marianum ITS1, 5.8S rRNA gene and ITS2
Silybum marianum var. marianum isolate MD2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosom..
Silybum marianum var. albiflorum isolate MD13 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribos..
Silybum marianum var. albiflorum isolate MD9 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit riboso...
0.01

[c] *A. rosea*


Dyera costulata(MK039536.1)
eudicots | 2 leaves
eudicots | 25 leaves
Condylocarpon isthmicum(MK039535.1)
eudicots and unknown | 4 leaves
eudicots | 3 leaves
eudicots | 9 leaves
eudicots | 4 leaves
eudicots and monocots | 4 leaves
eudicots | 4 leaves
eudicots | 5 leaves
Wrightia pubescens(KP092774.1)
eudicots | 2 leaves
eudicots | 36 leaves
0.1


Condylocarpon isthmicum 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 26S ribosomal RNA gene, partial se..
eudicots | 25 leaves
eudicots | 2 leaves
Dyera costulata internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 26S ribosomal RNA gene, partial sequence
eudicots | 36 leaves
eudicots | 2 leaves
Wrightia pubescens isolate SCBGP520_2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal tra...
eudicots | 5 leaves
eudicots | 4 leaves
eudicots and monocots | 4 leaves
eudicots | 4 leaves
eudicots | 9 leaves
eudicots | 3 leaves
eudicots and unknown | 4 leaves
0.02

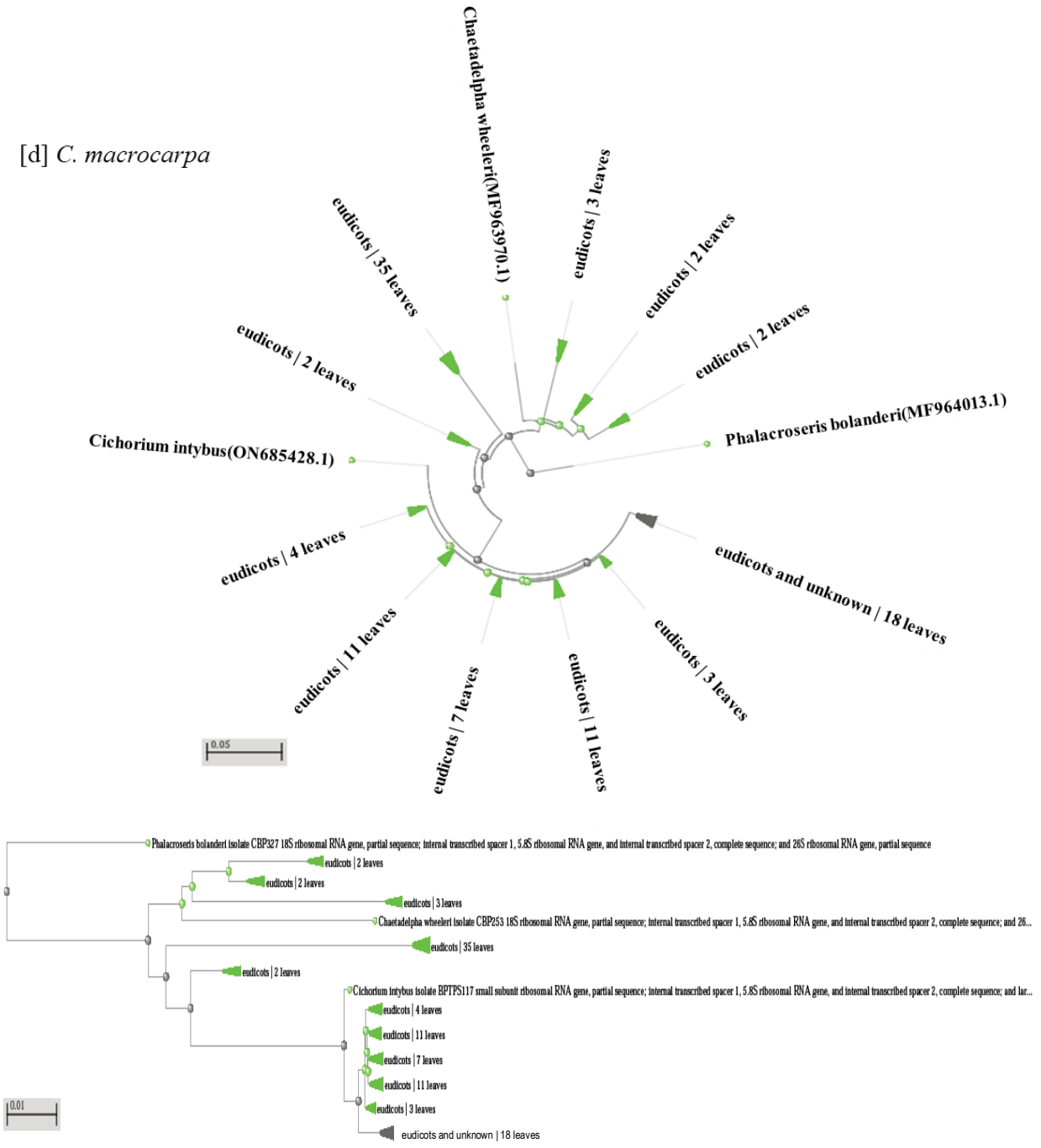


Figure 3: Distance trees of results for (a) *R. Regina*; (b) *D. Regina*; (c) *A. rosea*; (d) *C. macrocarpa*

Figure 4 displays the areas of clustering for the four plants which gave positive results based on the UniProtKB, when BLAST analysis alone was applied on the DNA ITS sequences. The selected ITS sequence of each plant is exceptionally similar, in terms of protein-coding sequences. Figure 5 shows the BLAST 250 results for the identity, score and value of *Roystonea regia*, *Delonix regia*, *Alcea rosea* and *Carissa macrocarpa*.

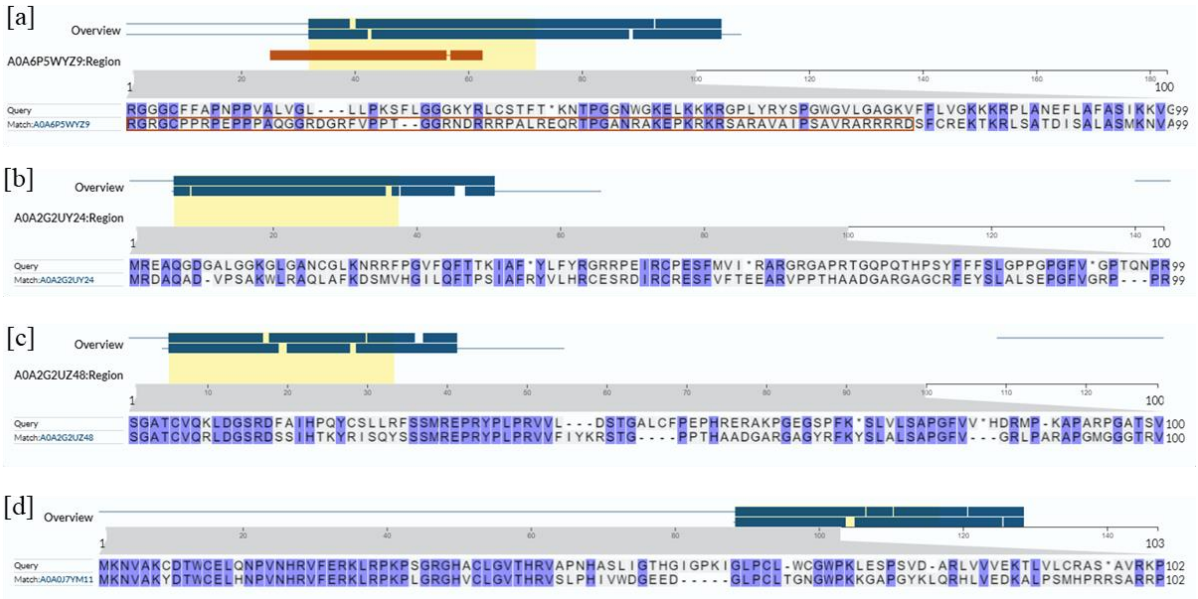


Figure 4: BLAST results found in UniProtKB (a) *Roystonea regia*, clustering was perceived at the A0A6P5WYZ9 area, (b) *Delonix regia*, clustering was perceived at the A0A2G2UY24 area, (c) *Alcea rosea*, clustering was perceived at the A0A2G2UZ48 area, and (d) *Carissa macrocarpa*, clustering was perceived at the A0A0J7YM11 area

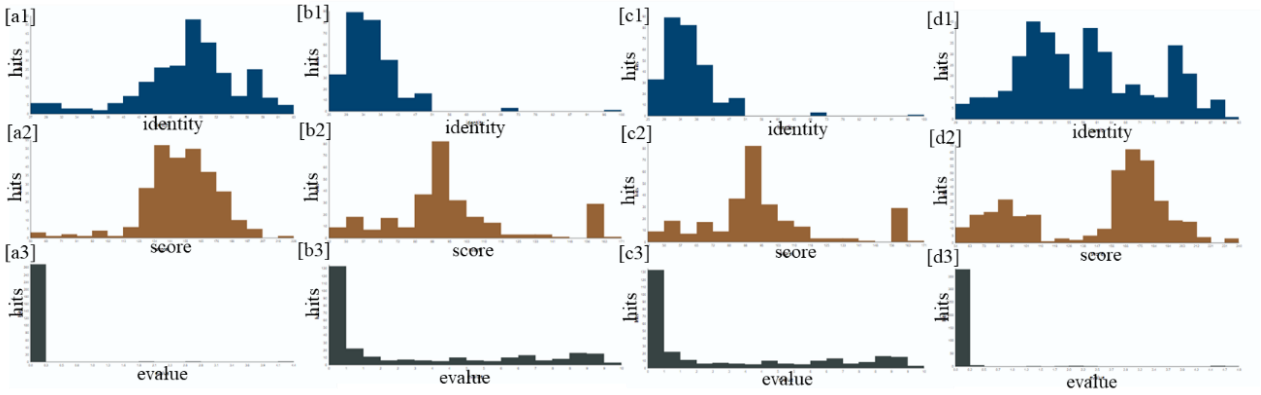


Figure 5: BLAST parameters for protein sequence from BLAST sequence similarity in UniProtKB. (a) BLAST 250 results for identity, score, and evalue for *Roystonea regia.* (b) BLAST 240 results for identity, score, and evalue for *Delonix regia* (c) BLAST 250 results for identity, score, and evalue have shown respectively for *Alcea rosea* (d) BLAST 250 results for identity, score, and evalue for *Carissa macrocarpa*

Discussion

Molecular markers facilitate investigations directed at scrutinising the genetic differences among medicinal plant populations (Yang et al. 2022). Research focusing on genetic variations, can contribute towards the identification and conservation of unique genotypes. As an example, the use of microsatellite markers, to assess the genetic diversity and population structure of *Panax ginseng*, contributed greatly towards the conservation of this highly valued medicinal herb (Lee et al. 2020).

Molecular markers can be relied on to provide an accurate assessment of medicinal plant species (Zhu et al. 2022). DNA barcoding, an approach that takes specific

molecular markers, is used for making distinctions between closely linked species. This is similar to what has been earlier observed, plant variety identification was achieved through the use of DNA barcoding (Chen et al. 2014).

In a study concentrating on the *Hypercium perforatum* species, population differentiation and gene flow dynamics were observed through the using of molecular markers (Crockett et al. 2004). This was supported by Solouki et al. (2008) who observed the identification of molecular markers associated with ideal features can support the efforts to propagate medicinal plants with exceptional qualities. Random amplified polymorphic DNA (RAPD) was helpful *Matricaria chamomilla* study to make the selection process easier of chamomile varieties with the highest content of essential oil.

Molecular markers can be used to trace the evolutionary development of medicinal plants. The use of markers in phylogenetic analyses, contributes towards a better understanding of the relationships between plant species. In a previous study, chloroplast DNA markers were applied, to distinguish the phylogenetic links, among a variety of *Artemisia* species (Kim et al. 2020).

A study, focusing on *Ginkgo biloba*, used dominant markers RAPD and amplified fragment length polymorphism (AFLP), and the genetic diversity index (H), to investigate the genetic basis of ginkgolide content disparity, among different populations (Zhou et al. 2020). The capacity of molecular markers, to precisely distinguish the genetic relationship between plant species, is partly due to the fact that unlike morphological markers, molecular markers are not influenced by external environmental factors (Hamouda 2019). Molecular markers can be used to acquire important information regarding the phylogenetic linkage between the various accessions, and can also prove to be useful for the generation of diagnostic fingerprints (Ahmad 2021). The availability of useful information, regarding tissue and plant type, will facilitate the gathering of appropriate plant materials, for purposes including propagation, research and curative treatments.

Conclusion

Molecular markers can be used not only for ensuring the quality and consistency of medicinal plant products, but also for the verification of product authenticity, through the identification of markers associated with specific medicinal compounds. To conclude, information derived through molecular markers, regarding the genetic relationships, diversity and traits of medicinal plants, will contribute towards efforts directed at their conservation, propagation and quality preservation.

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Conflicts of interest

There are no conflicts of interest to disclose.

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